

Algebra II Final Targeted Practice

Instructions

Complete in 60 minutes. Show clear work. Use exact values unless a decimal approximation is explicitly requested. For triangle measurements, round side lengths to the nearest tenth and angles to the nearest degree unless exact values are possible.

Reference Facts

- Principal inverse trig ranges:

$$\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \cos^{-1} x \in [0, \pi], \quad \tan^{-1} x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

- Law of Sines and Law of Cosines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}, \quad c^2 = a^2 + b^2 - 2ab \cos C$$

- Triangle area:

$$K = \frac{1}{2}ab \sin C$$

- Conic forms:

$$\begin{aligned} (x-h)^2 + (y-k)^2 &= r^2, & (x-h)^2 &= 4p(y-k), & (y-k)^2 &= 4p(x-h) \\ \frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} &= 1, & \frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} &= 1, & \frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} &= 1 \end{aligned}$$

Problems

1. Find the exact value of each expression.

$$\sec\left(\tan^{-1}\left(-\frac{5}{12}\right)\right), \quad \sin\left(\cos^{-1}\left(-\frac{7}{25}\right)\right), \quad \cot\left(\sin^{-1}\left(\frac{4}{5}\right)\right)$$

2. Solve on $0 \leq \theta < 2\pi$.

$$2 \cos^2 \theta - \cos \theta - 1 = 0$$

3. Solve on $0 \leq \theta < 2\pi$.

$$\sqrt{3} \sec \theta + 2 = 0$$

4. For

$$y = -3 \cos \left(2x + \frac{\pi}{2} \right) - 1,$$

state the amplitude, period, phase shift, midline, and range. Then give one interval that contains exactly one full period.

5. Write an equation of a sine function with amplitude 2, midline $y = -3$, period 4π , and passing through the point $(\frac{\pi}{2}, -3)$ at the beginning of a rising cycle.
6. In $\triangle ABC$, $a = 16$, $b = 21$, and $C = 58^\circ$. Find c , the area of the triangle, and the larger of $\angle A$ and $\angle B$.
7. In $\triangle PQR$, $p = 18$, $q = 24$, and $P = 42^\circ$. Determine how many triangles are possible. Solve every possible triangle.
8. A hiker walks 5.2 miles on a bearing of 038° , then 7.1 miles on a bearing of 128° . How far is the hiker from the starting point?

9. Identify the conic and rewrite in standard form.

$$x^2 - 10x + 8y - 7 = 0$$

Then state its vertex, focus, directrix, and axis of symmetry.

10. Identify the conic and rewrite in standard form.

$$9x^2 + 4y^2 - 54x + 32y + 109 = 0$$

Then state the center and the lengths of the major and minor axes.

11. Identify the conic and rewrite in standard form.

$$25x^2 - 16y^2 - 200x - 32y - 16 = 0$$

Then state the center, vertices, foci, and asymptotes.

12. Write an equation of the hyperbola with center $(-2, 3)$, vertices $(-2, 8)$ and $(-2, -2)$, and asymptotes

$$y - 3 = \pm \frac{5}{4}(x + 2).$$

Then find the coordinates of the foci.